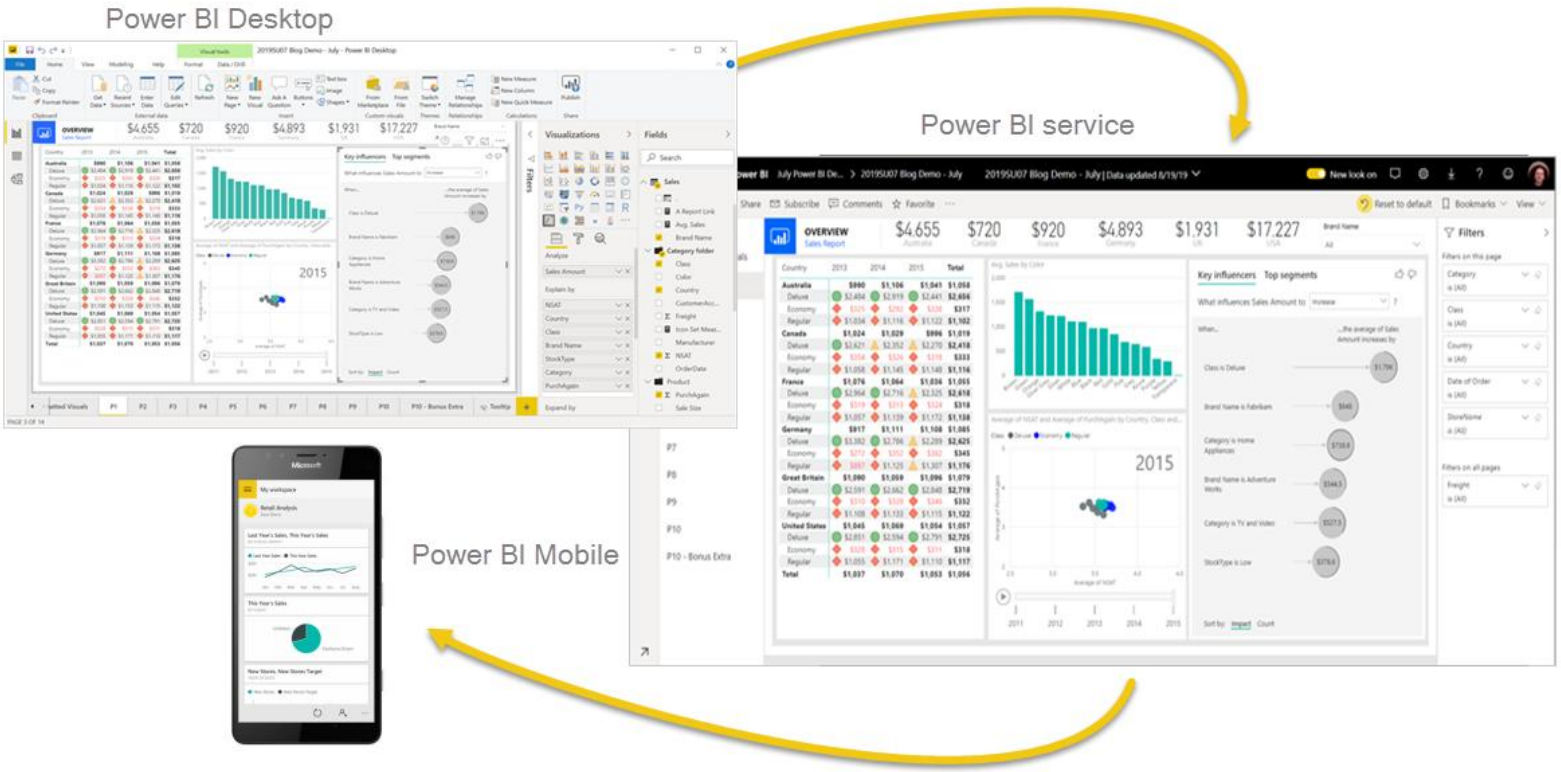
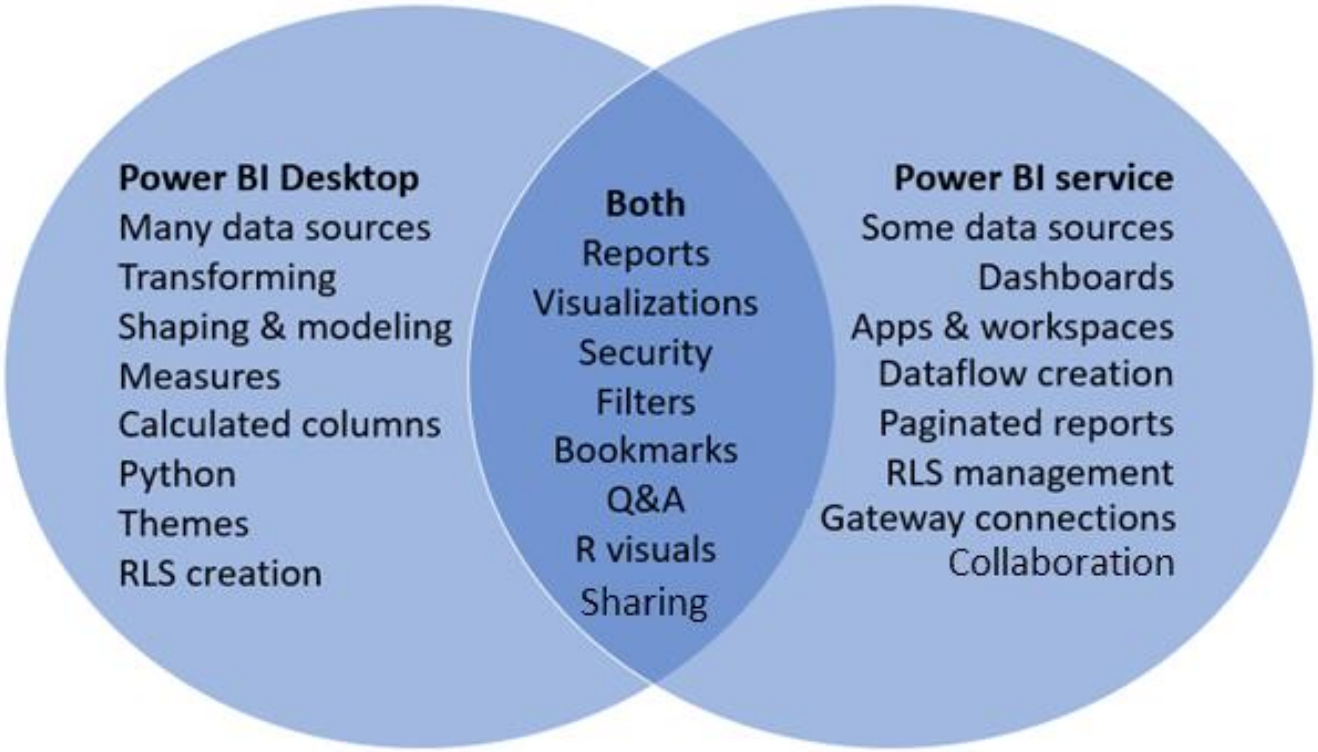
# Microsoft Power BI Products



# Power BI Desktop vs Power BI Service



# Power BI Licensing Models

- Power BI Desktop and Power BI Service Fabric Free
  - Supporting single users
- Power BI Pro
  - For Power BI Service
  - Sharing data, dashboards, reports, etc., among multiple users for team collaboration
- Power BI Premium
  - For Power BI Service
  - Supporting dedicated Power BI Report Servers
  - 60-day free trial

    https://powerbi.microsoft.com/en-us/get-started/

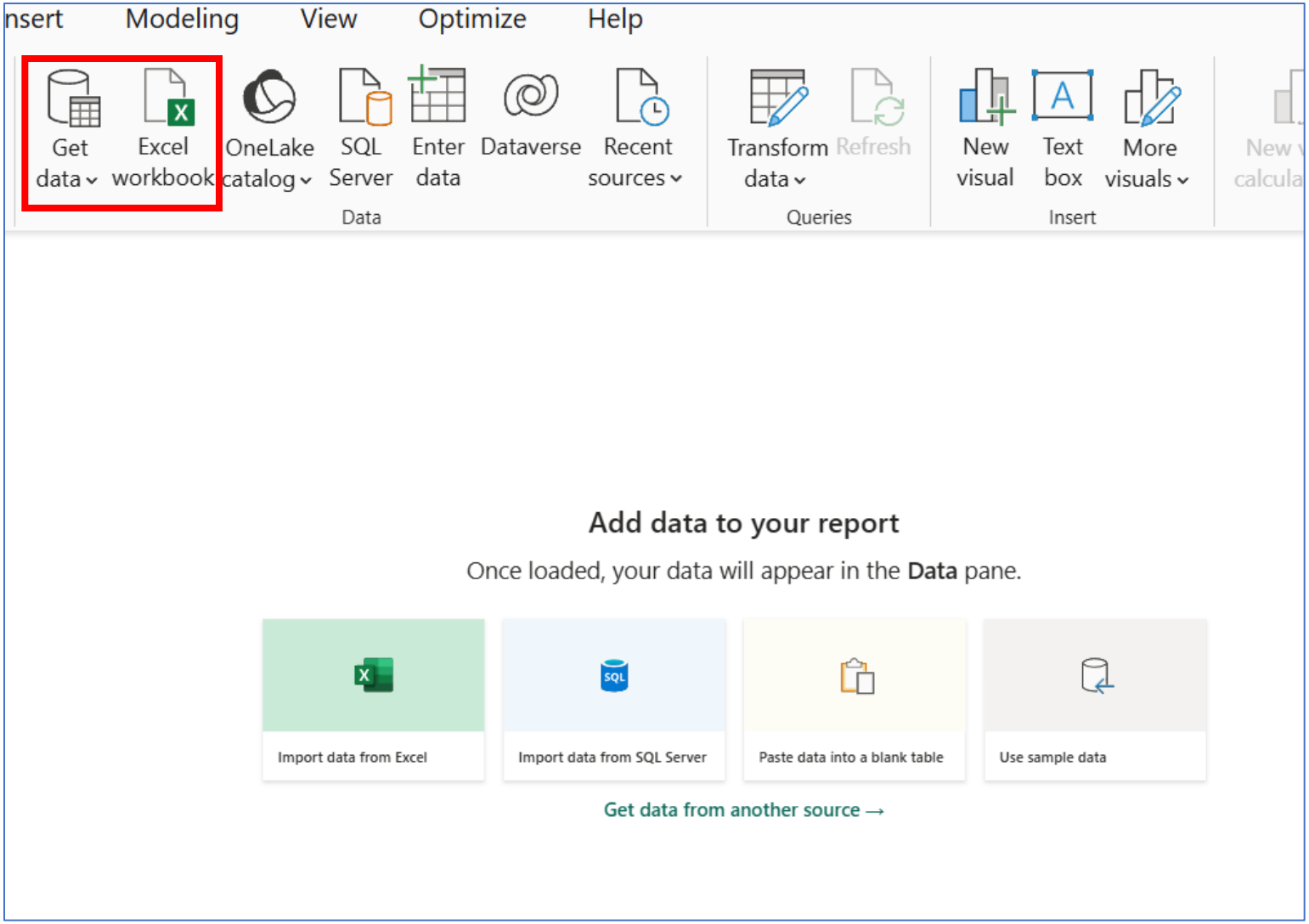
# Power BI Objects

- Visuals
  - A visualization (or visual) is a type of chart for displaying the data from datasets
  - Visuals are interactive
  - A *tile* is a rectangular box that holds an individual visual
- Datasets
  - A dataset is a container of data
- Dashboards
  - A dashboard is a single screen with visuals and text to facilitate business decision-making
  - Generally, a dashboard addresses a central area of interest or answers a single question
  - A dashboard can be based on multiple datasets
    - Current limitations: the interactive visuals in dashboards may be much slower than those in reports
  - Current limitation: Power BI Desktop does not support dashboards
- Reports
  - A report is one or more pages of visuals and text
  - A report is based only on a single dataset
    - Multiple data sources must be combined into a single dataset first
  - Generally, each page in a report addresses a central area of interest or answers a single question

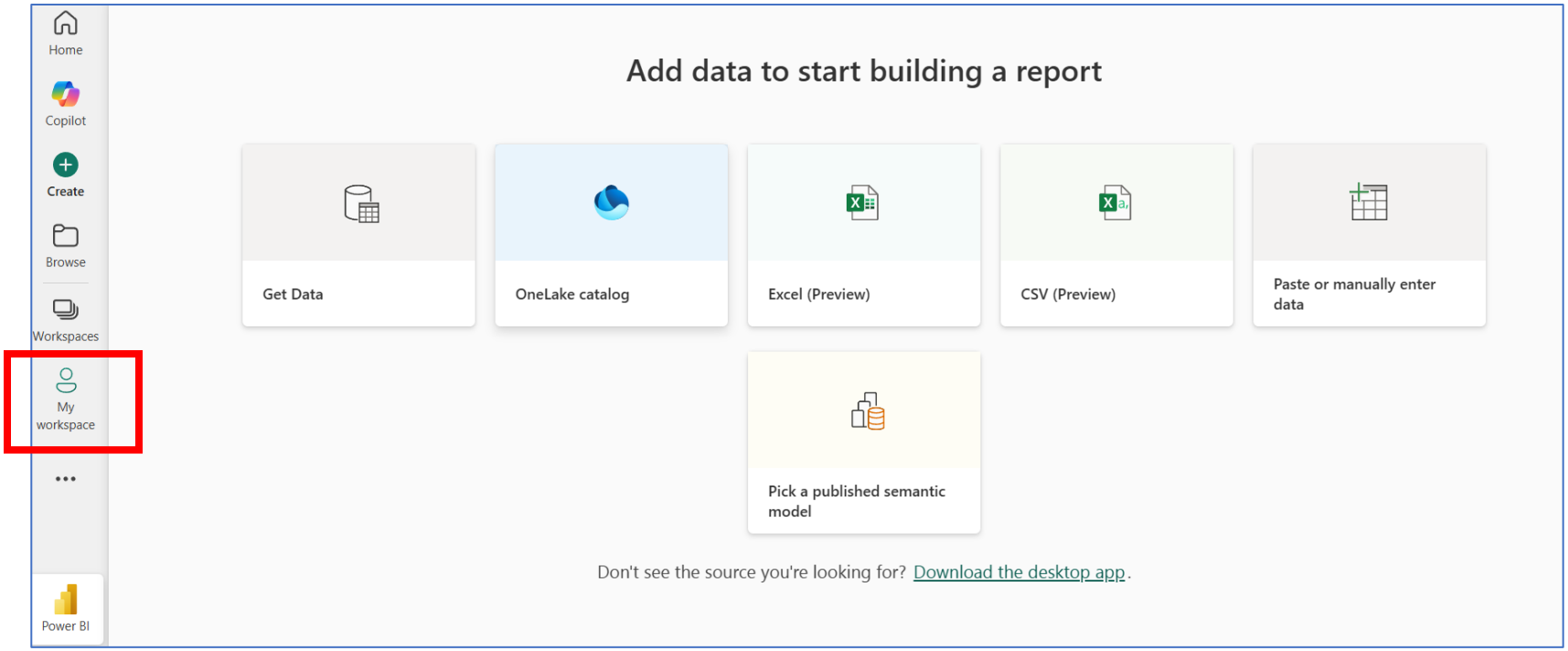
# Power BI User Roles

- Business Users
- Dashboard/Report Creators
- Administrators
- Advanced Developers
  - Creating own types of visuals
  - Creating Power BI embedded analytics solutions (i.e., embedding Power BI objects in a web application)
  - Using the Power BI API in other software
  - Integrating Power BI with other analytics tools

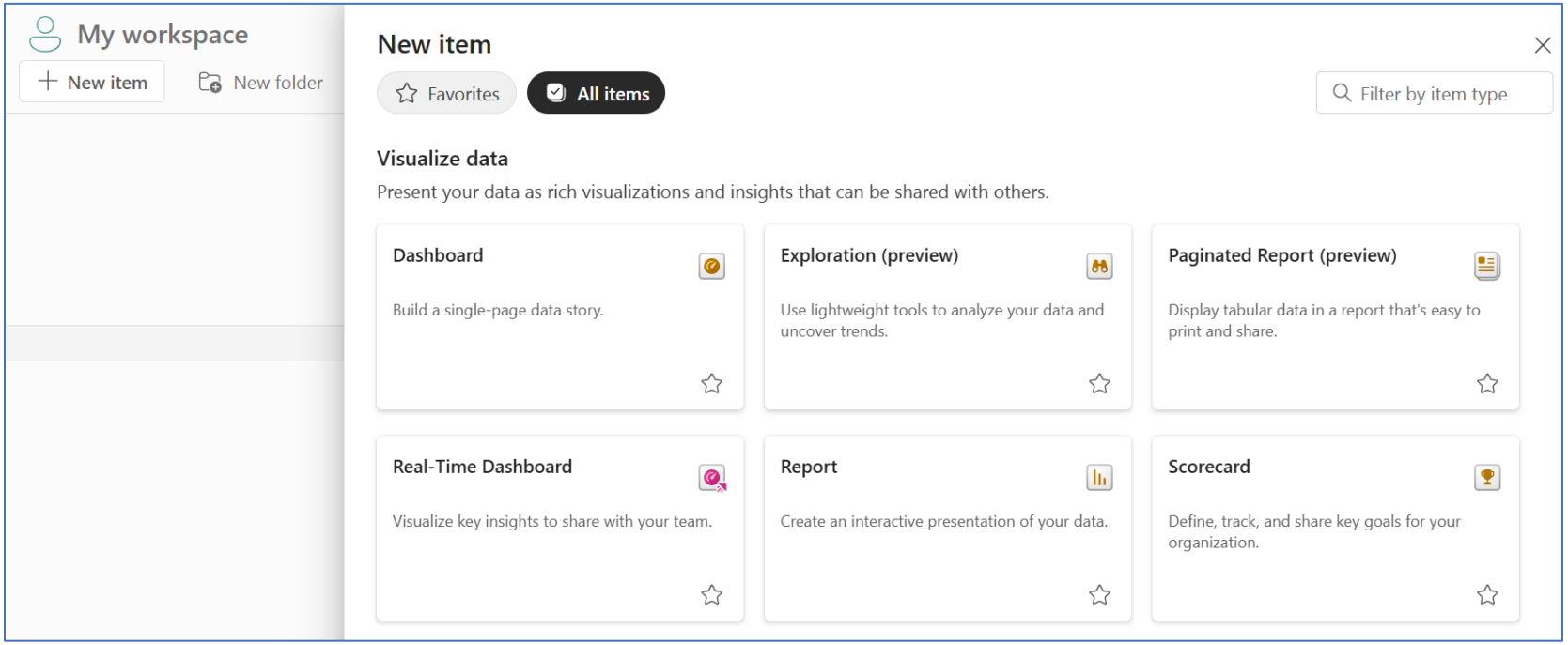
# Data Sources on Power BI Desktop



# Data Sources on Power BI Service



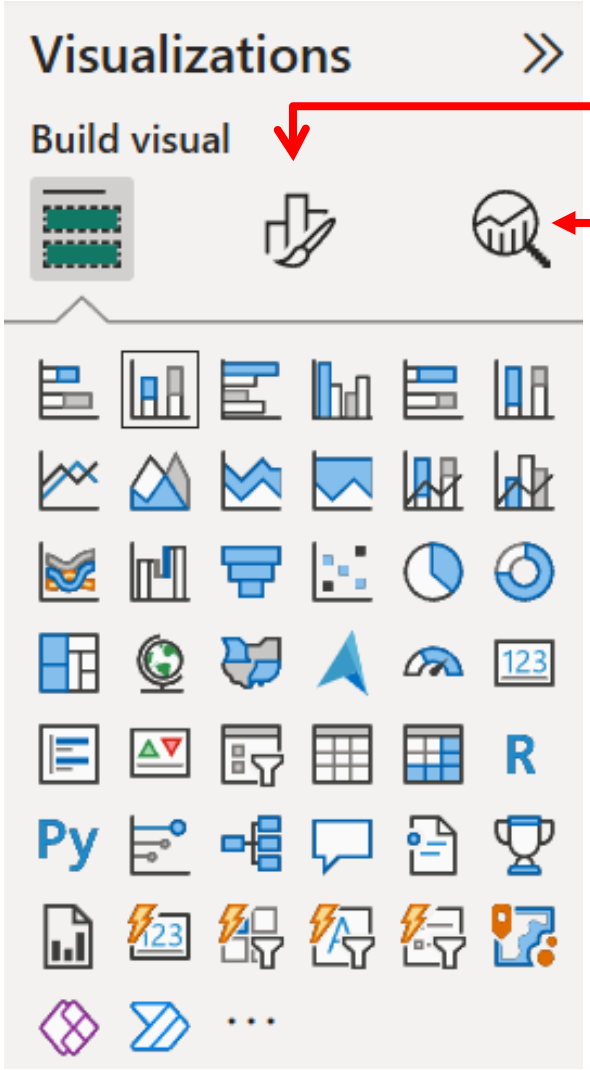
# Creating a Report in Power BI Service



# Two Modes of Reports in Power BI Service

- Two modes for interacting with reports
  - Editing view
    - Used by report creators
  - Reading view
    - Used by business users
- If you are the owner of a report, you can freely switch one view mode to the other

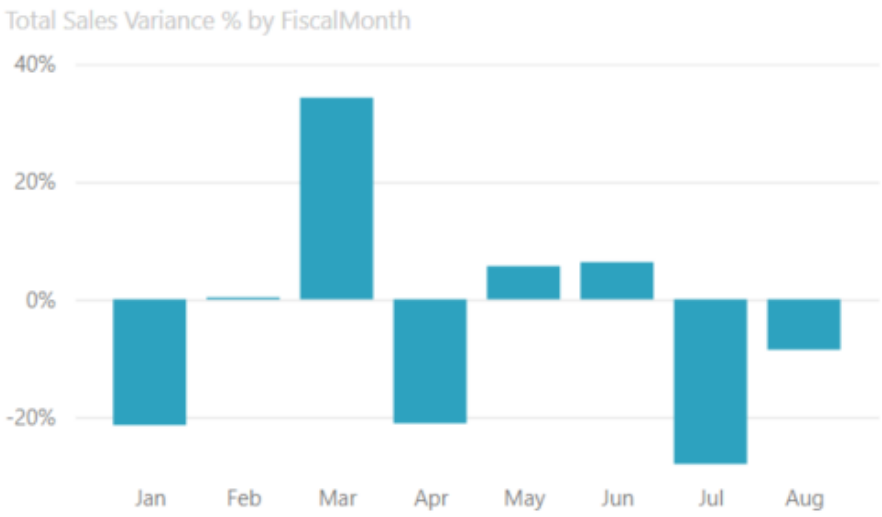
# Power BI Visuals

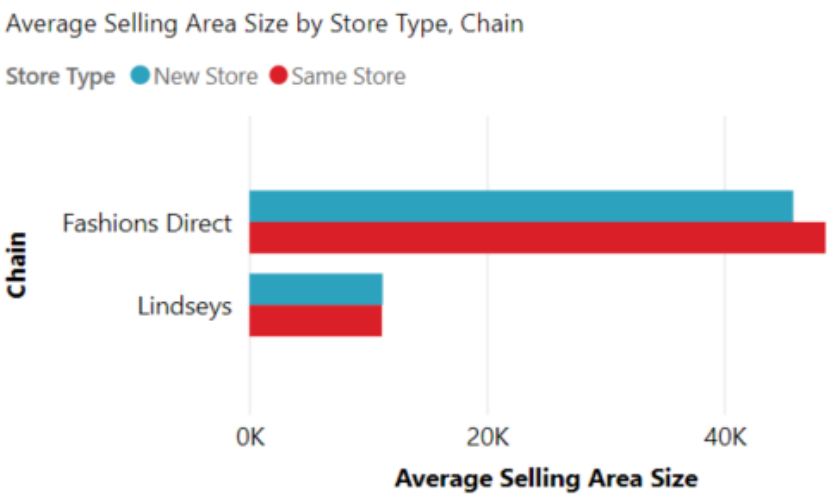



Formatting

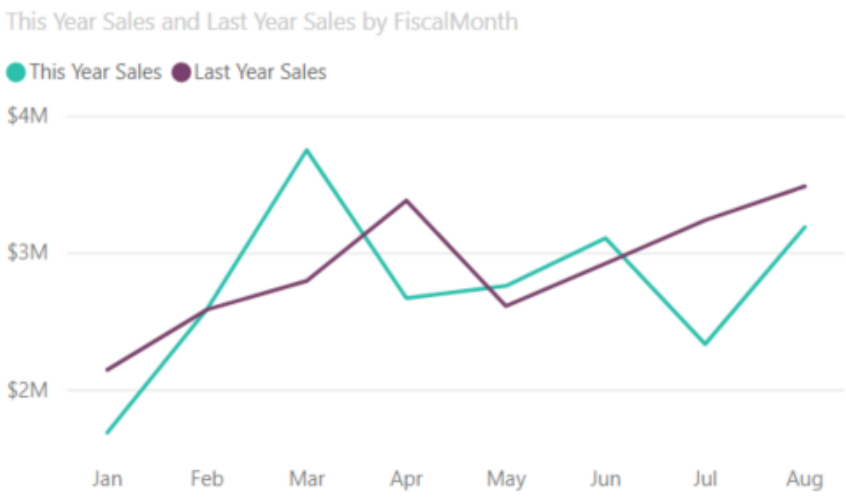
Benchmarking

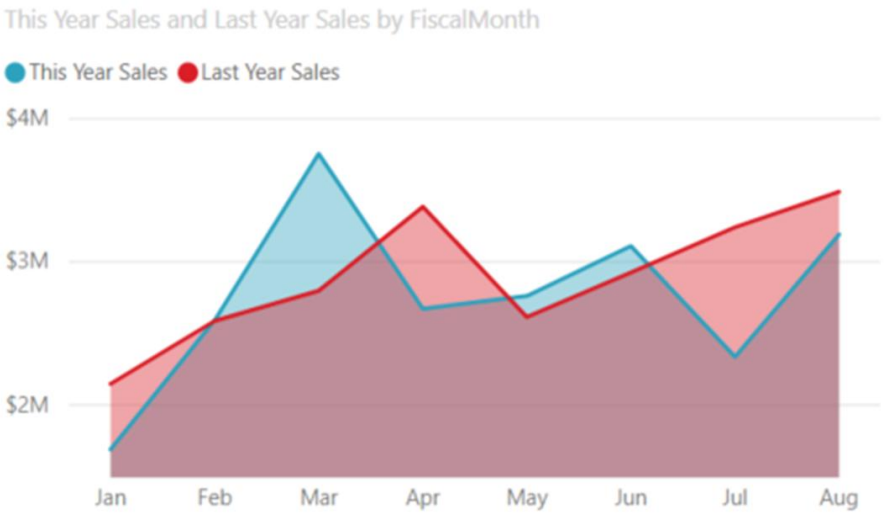
# Bar and Column Charts



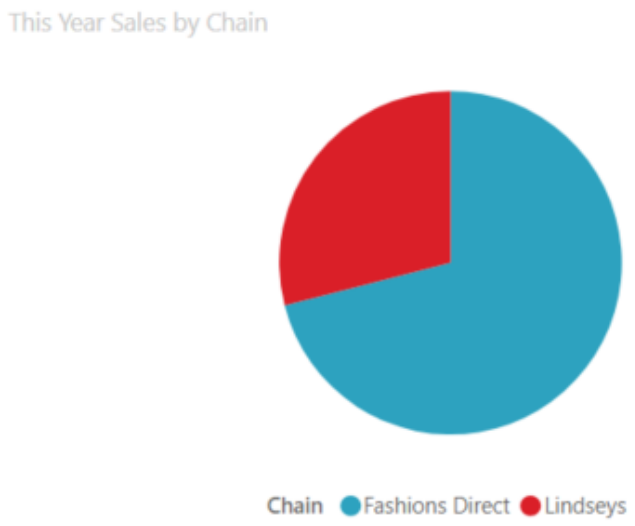
# Line and Area Charts

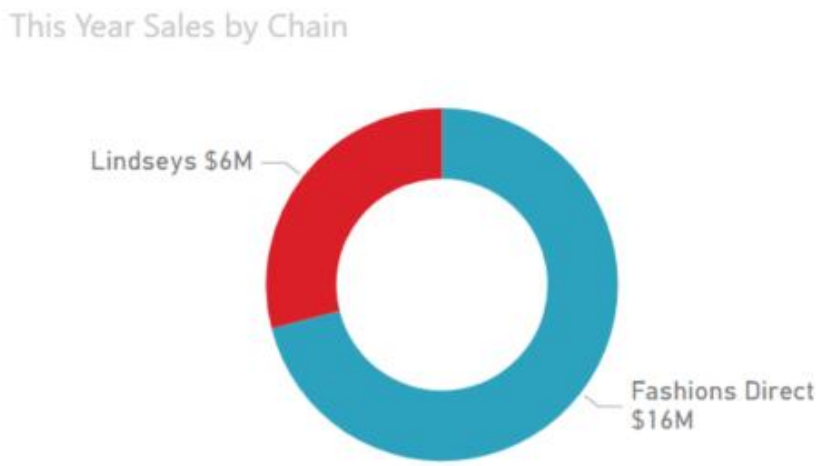


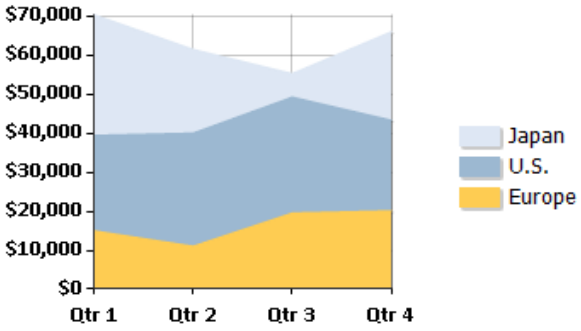
# Combo Charts

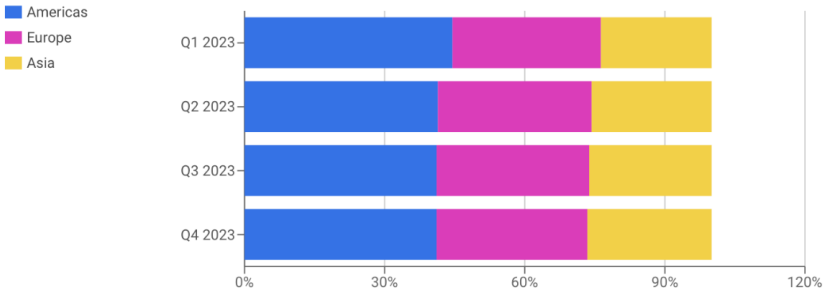
# Pie and Donut Charts

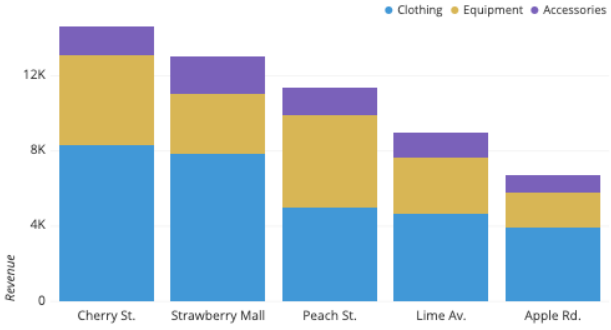


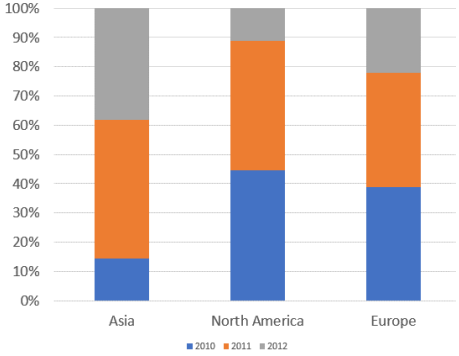



# Stacked Charts

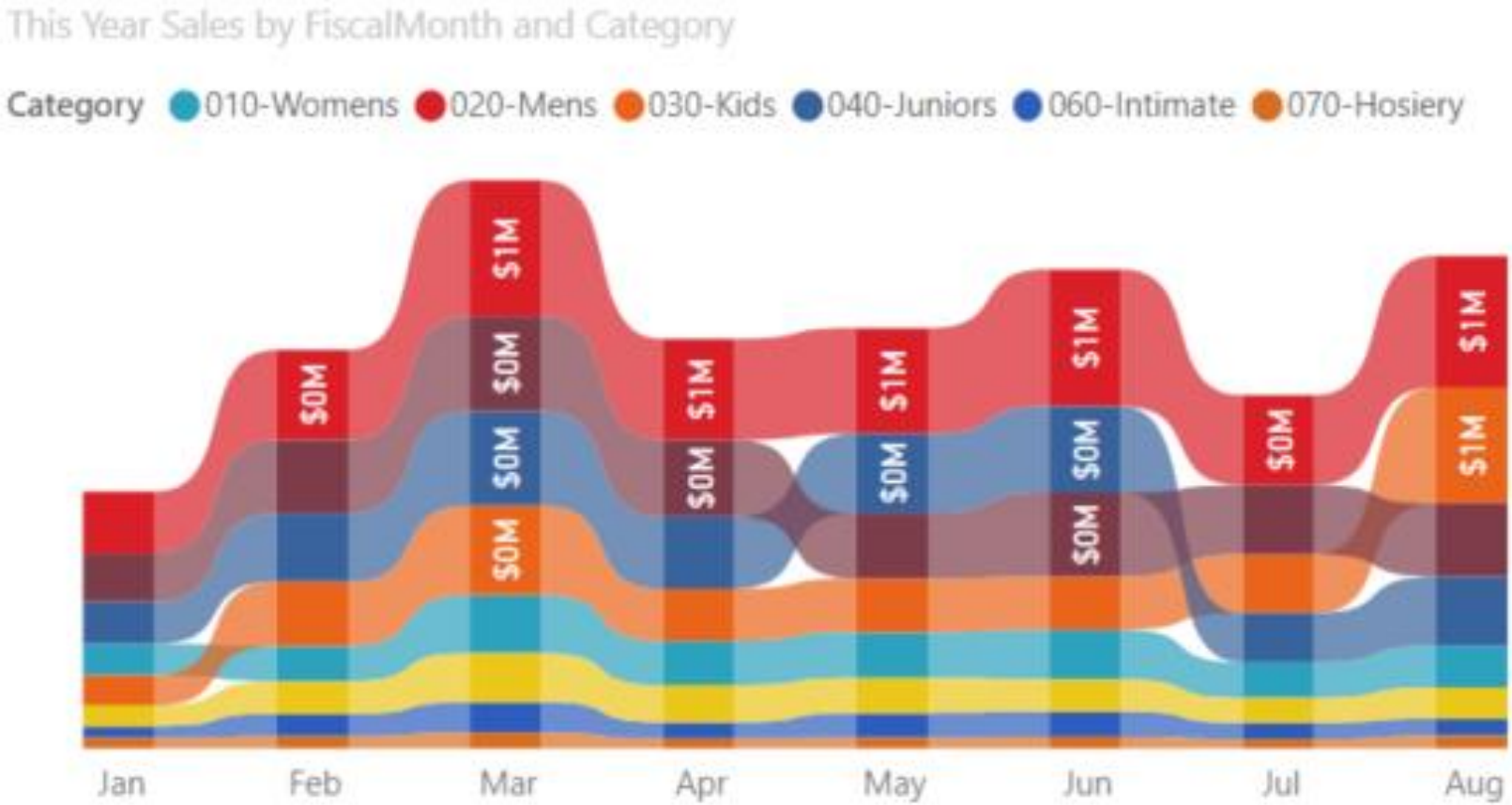


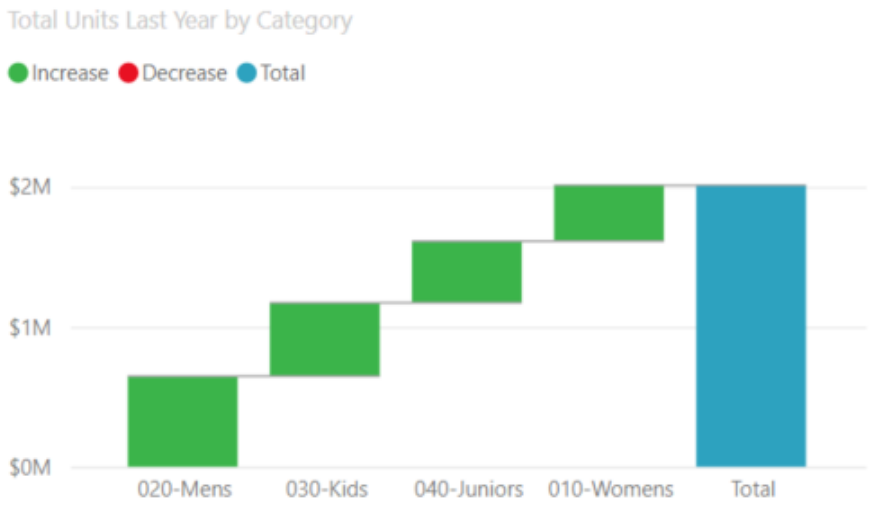



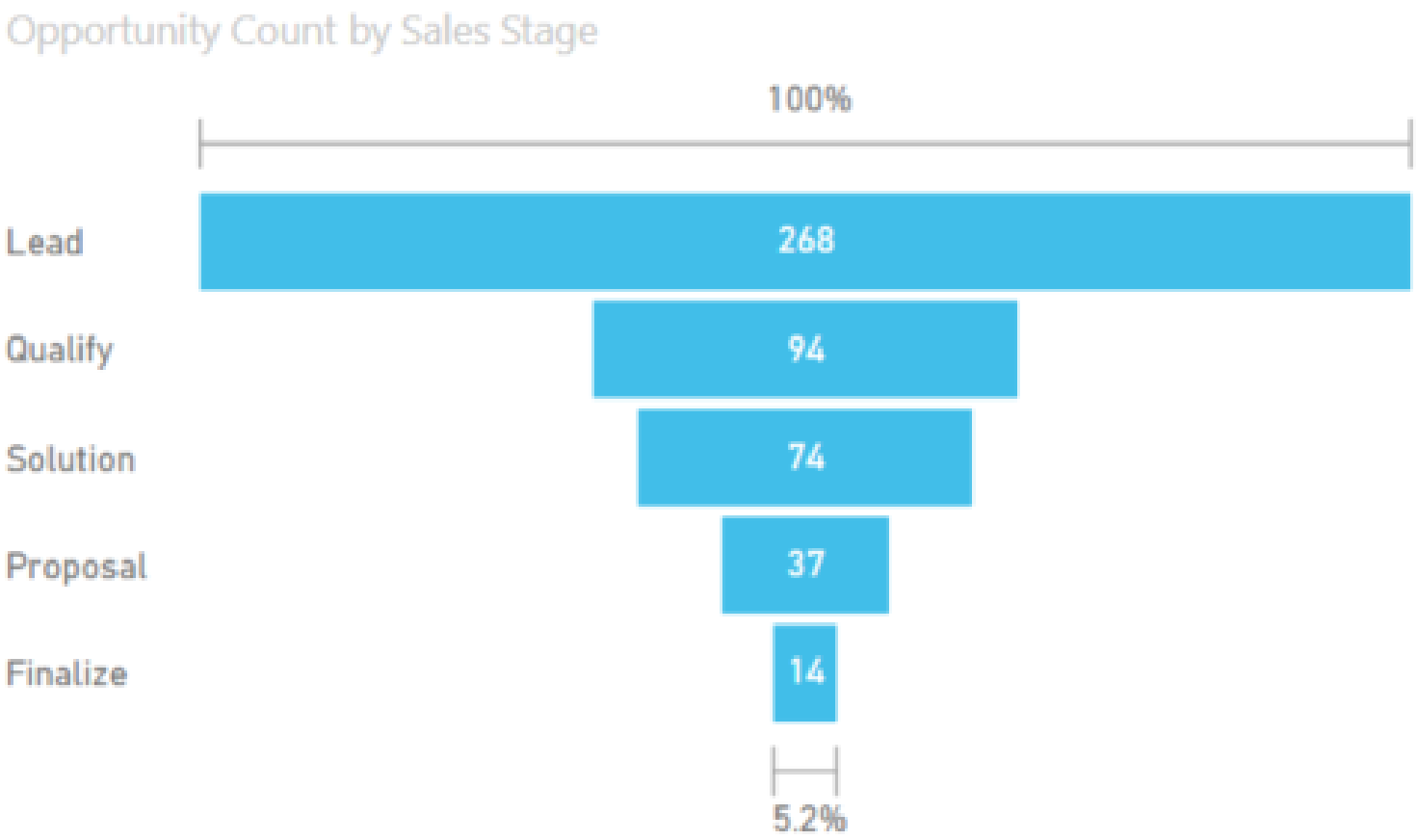
# Ribbon Charts



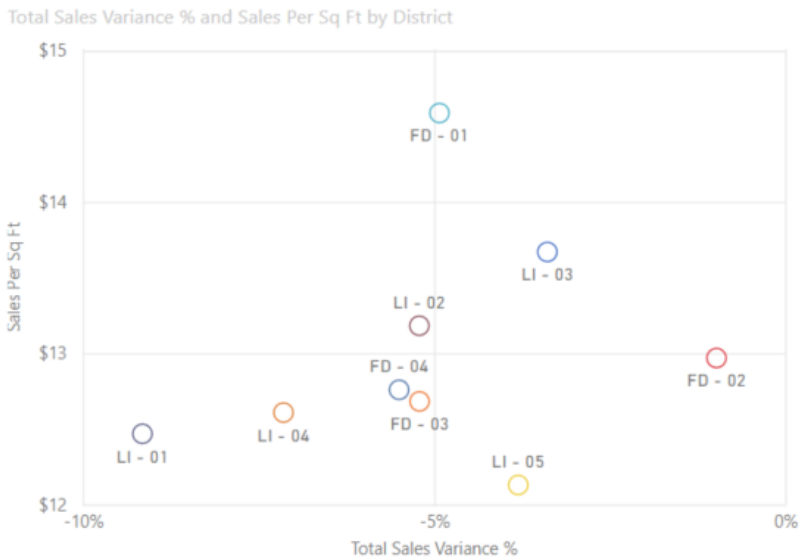
# Waterfall Charts

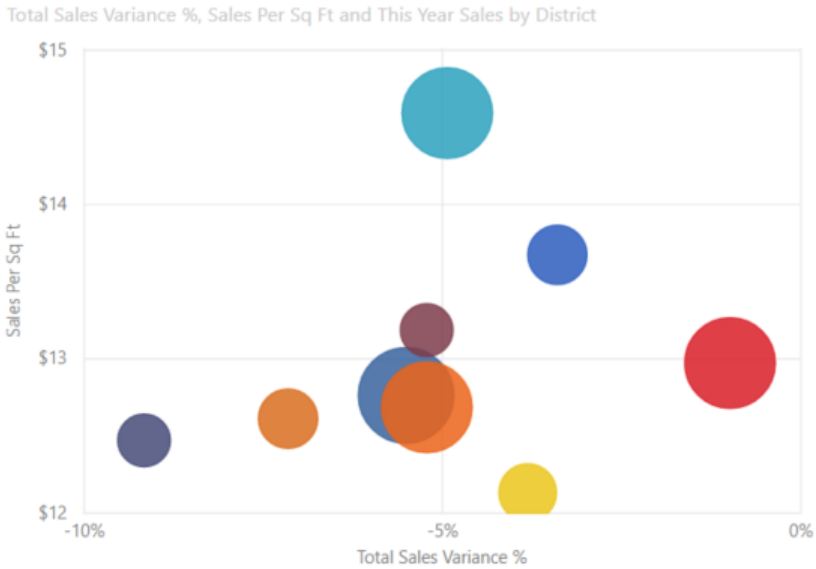


# Funnel Charts

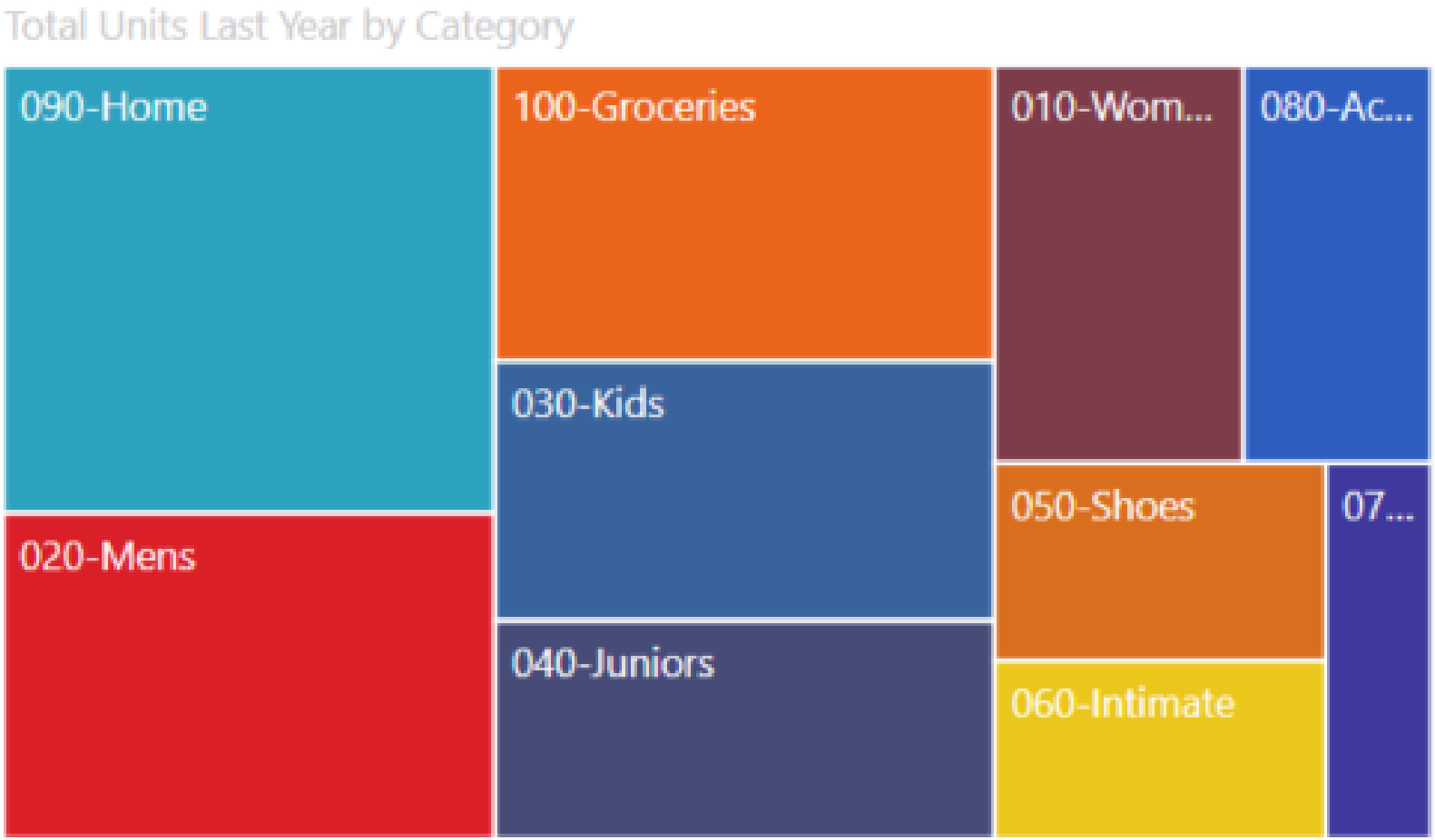


# Scatter and Bubble Charts

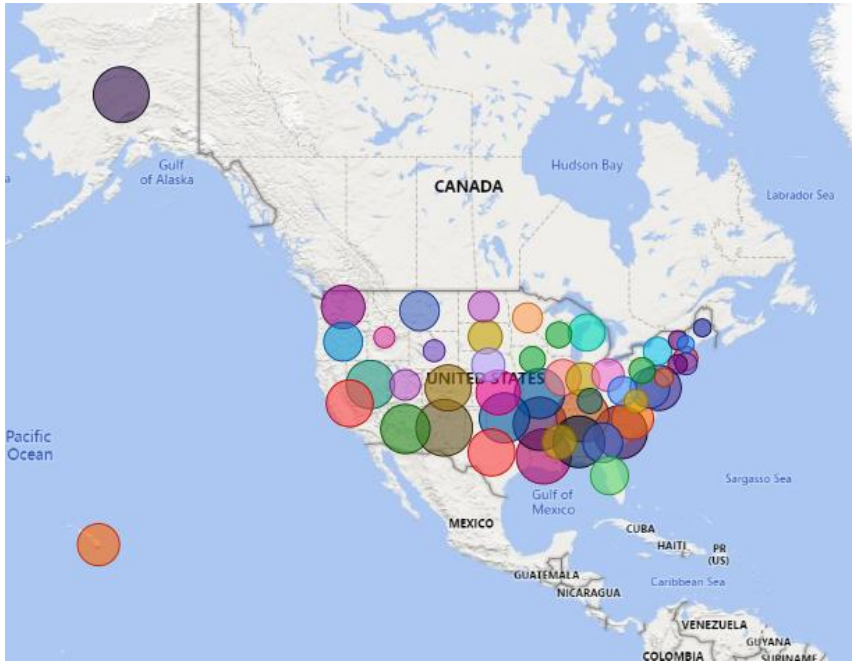


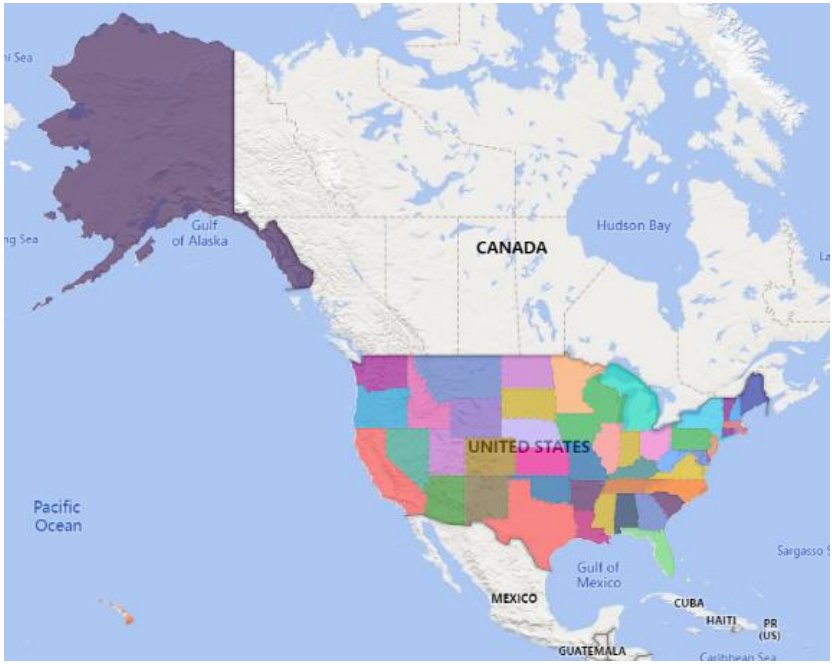



# Treemaps

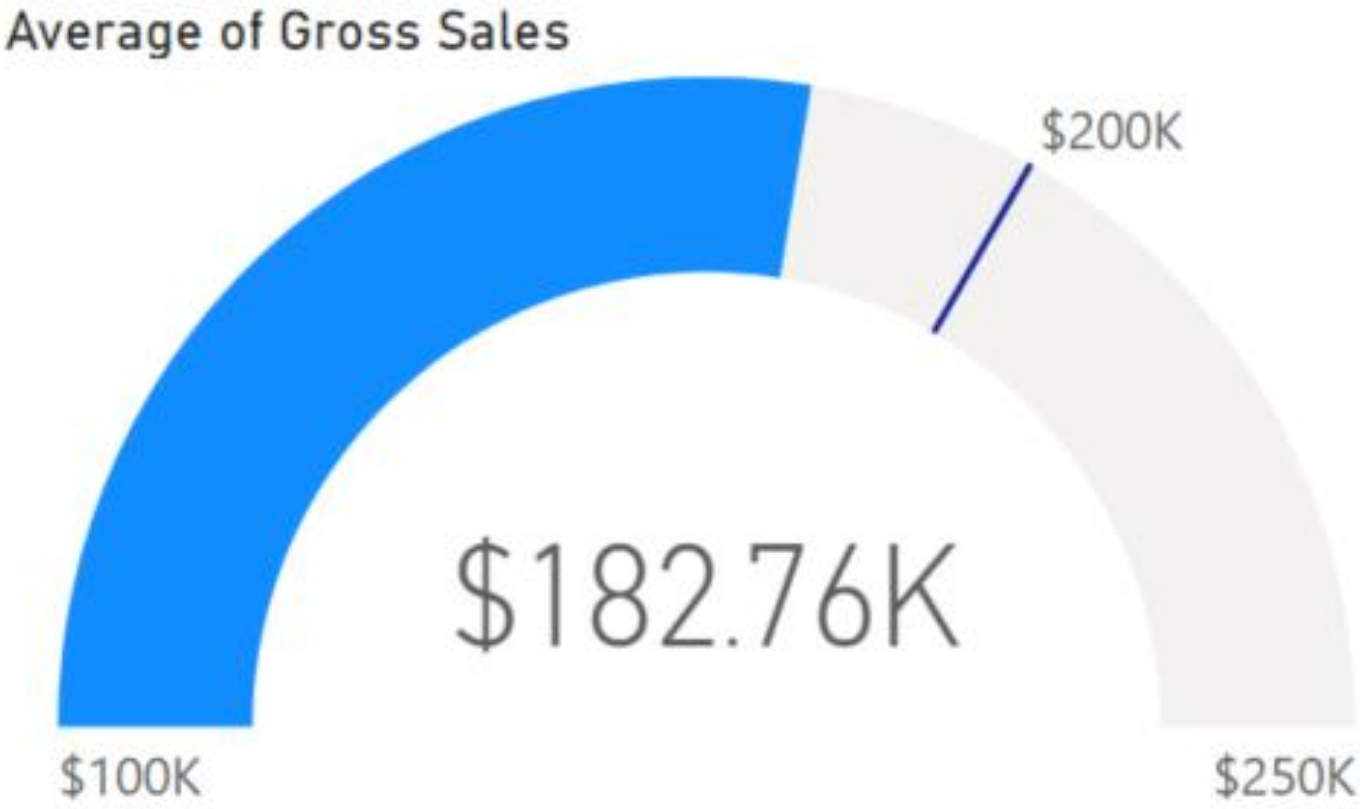

Total Units Last Year by Category
090-Home
100-Groceries
010-Wom...
080-Ac...
030-Kids
050-Shoes
07...
020-Mens
040-Juniors
060-Intimate

# Maps and Filled Maps



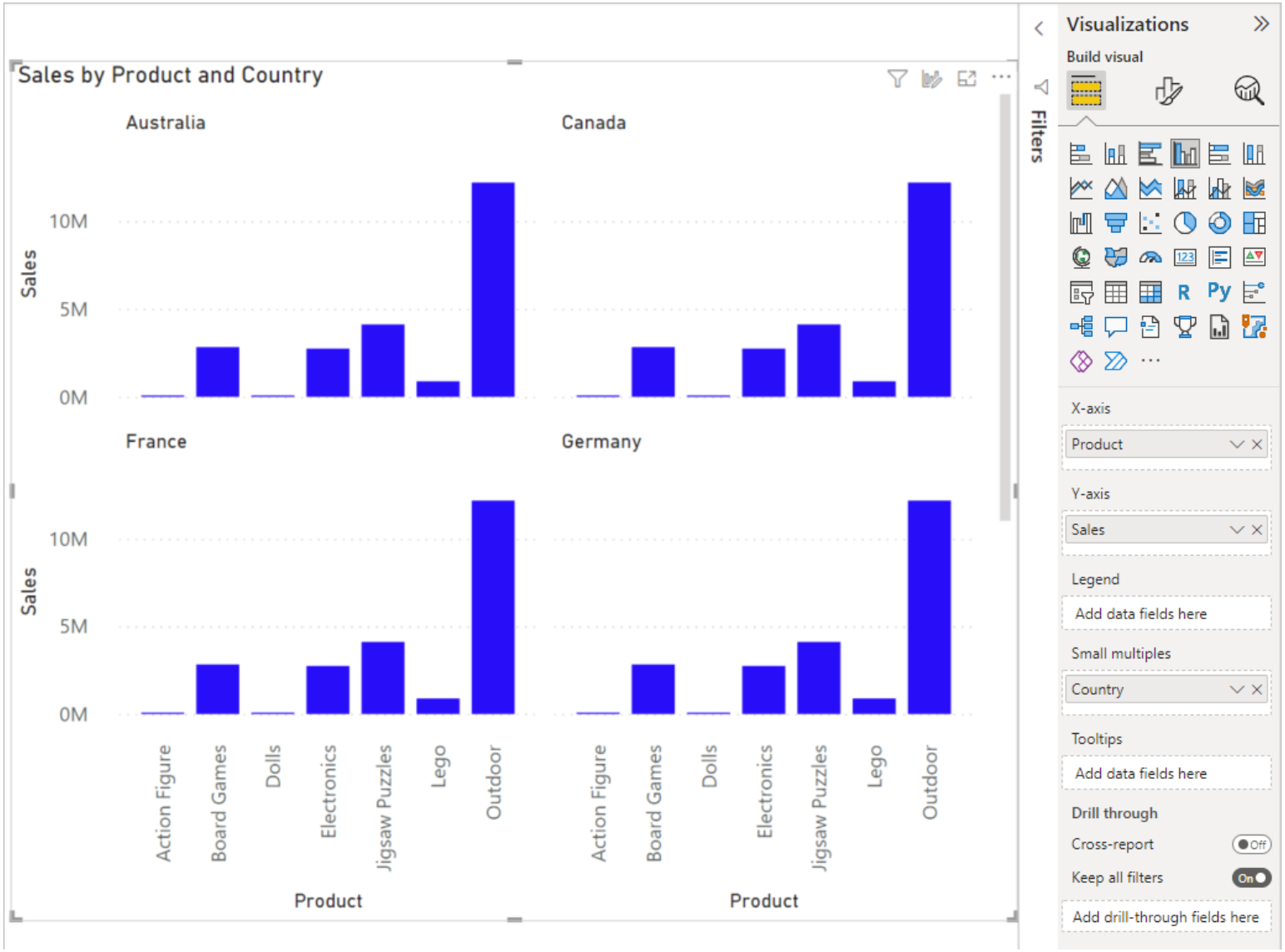
# Radial Gauge Charts



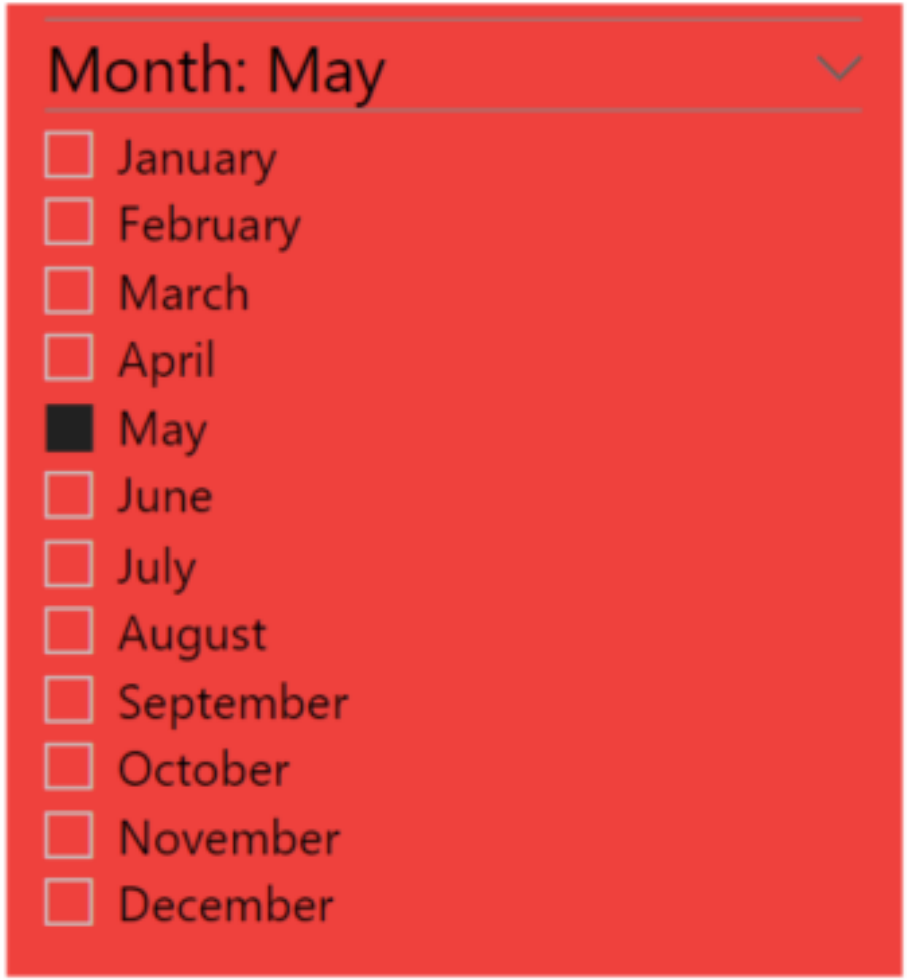
# KPI Charts

Total Units This Year and Total Units Last Year by Month
$482,537
Goal: $595,092 (-18.91%)

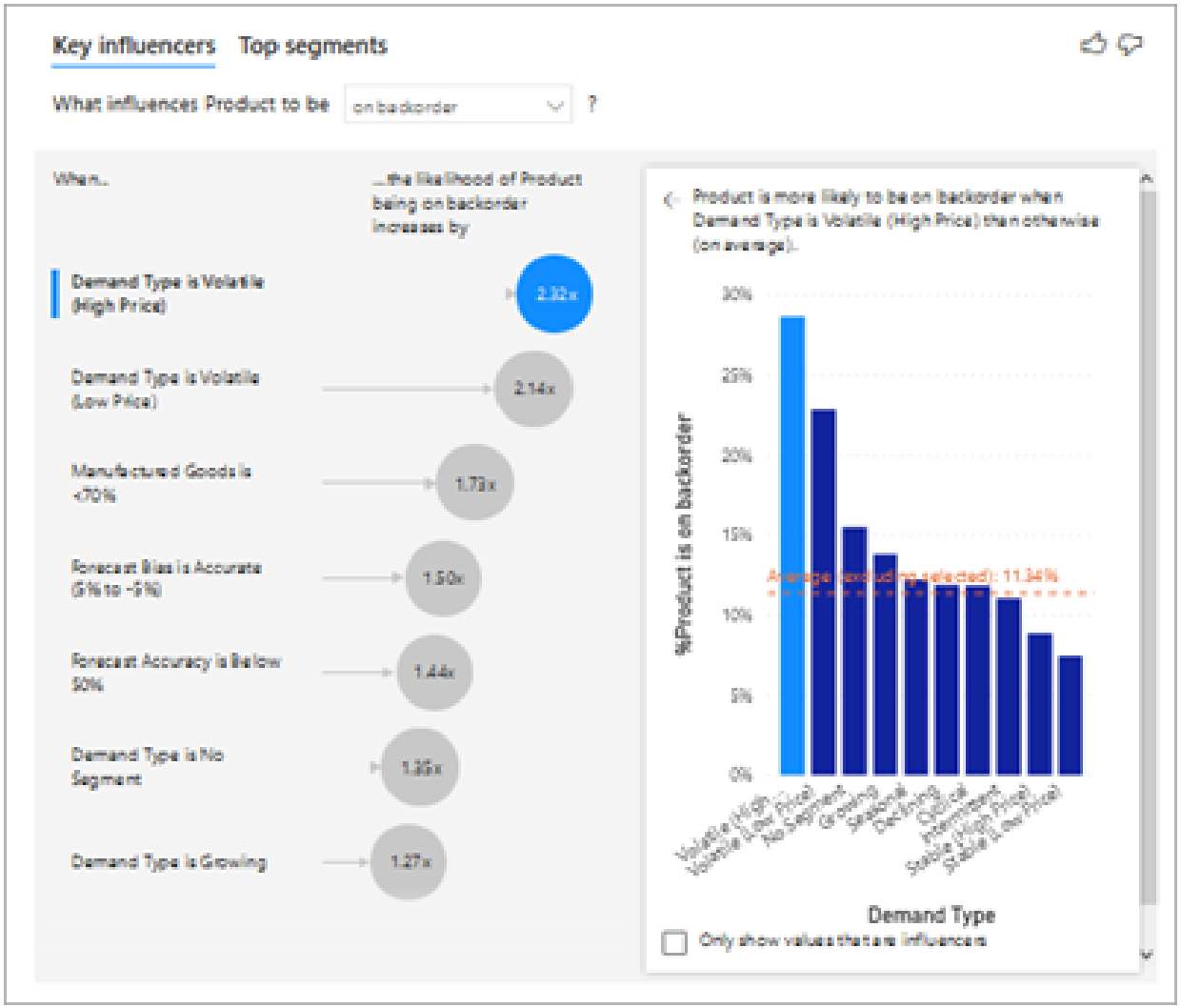
# Small Multiples



# Slicer

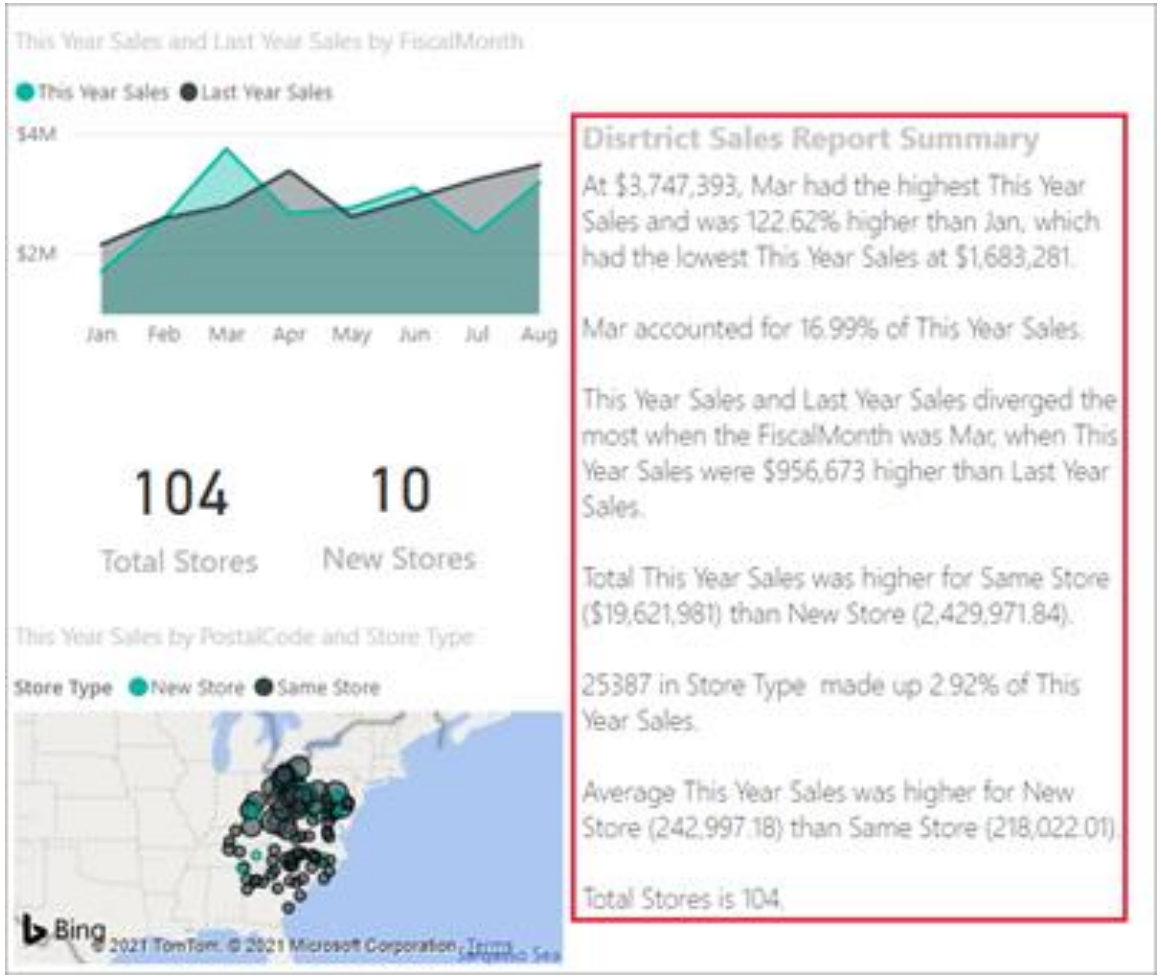

Month: May
January
February
March
April
May
June
July
August
September
October
November
December

# Key Influencers

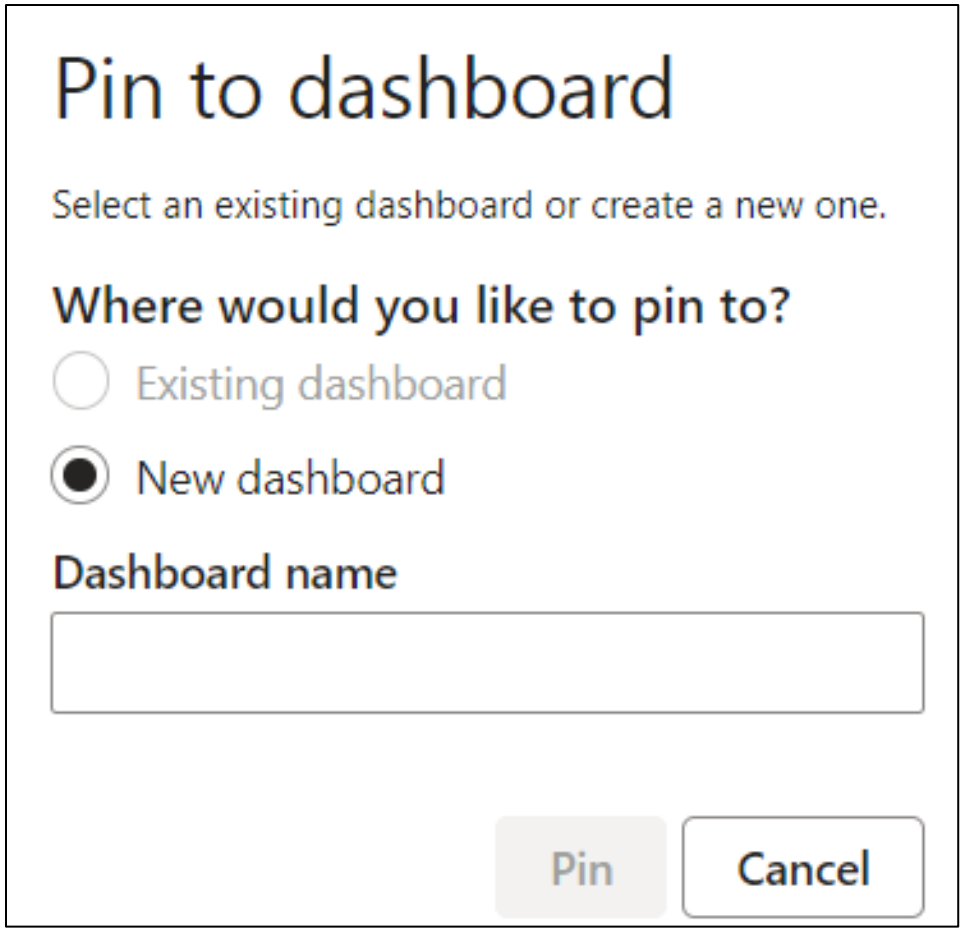


# Decomposition Tree

# Smart Narrative



# Creating a Dashboard from Reports



# Report/Dashboard Design

- Report/dashboard design
  - The fundamental challenge of report/dashboard design is to display all the required information clearly and without distraction, in a manner that can be assimilated quickly
- Three layer of information
  - Monitoring
  - Analysis
  - Management

# Design Principles

- Use of visuals to highlight data
- Transparent to business users, meaning that they require minimal training and are extremely easy to use
- Combining data from a variety of sources into a single, summarized, and unified view
- Presenting a dynamic, real-world view with timely data
- Requiring little coding or no coding to implement, deploy, and maintain

# Key Performance Indicators

- Outcome KPI
  - Lagging indicators, e.g., revenues
- Driver KPI
  - Leading indicators, e.g., sales leads
- Unique features of KPIs
  - Strategy
  - Targets
  - Ranges
  - Encodings
  - Time frames
  - Benchmarks